

RADAR Sleep Hygiene Study

YOU WILL BE GIVEN A COPY OF THIS INFORMATION SHEET

Ethical Clearance Reference Number: LRS/RGO-24/25-48044

Participant Information Sheet for RADAR Sleep Hygiene Study

You are invited to take part in this study about measuring sleep, mood, and other aspects of everyday life in healthy adults. Before deciding whether to participate, please take time to read this information sheet carefully and to discuss it with others if you wish. Please feel free to ask us (by email) if there is anything that is not clear or if you would like more information. If you decide to take part in this study, you will be asked to sign a consent form.

1. What is the purpose of this research?

The purpose of this project is to investigate the impact of smartphone use habits on sleep hygiene, that is, the impact of different types of phone interactions before bedtime on the quality, length, and effectiveness of sleep. Based on passive smartphone data, wearable sleep tracking, and self-reported behaviour, the investigation aims to identify patterns of digital use leading to healthy or disrupted sleep.

2. Who funds this research?

This study is funded by King's College London.

3. Why have I been invited to take part?

We have already collected data from autistic participants as part of the AIMS-2-TRIALS programme, and from patients with depression, epilepsy, multiple sclerosis and other conditions as part of the RADAR-CNS programme. You have been invited to take part in this study because we are interested conducting a population science study by collecting remote measurement data to examine the effects of smart phone usage habits on sleep. You are likely to be eligible based on your responses to our screening questions.

4. Do I have to take part?

No, it is your choice. You do not have to take part, and if you decide to participate you are **free to withdraw** from the study at any time, without giving a reason. If you are a King's College London employee or student, whether or not you decide to take part will not affect your staff/student experience, opportunities, or grades in any way.

5. What will happen if I take part?

The RADAR Sleep Hygiene Study involves two main parts. If you choose to take part, firstly, we will invite you to wear your wearable device as much as you can for at least three months, and as long as you are happy to, up to one year (we will refer to this as your Participation Period). Secondly, we will invite you to install two apps on your smartphone. One of these apps will enable you to report regularly on things like your sleep, mood, attention and behaviour. The other app will collect information from your smartphone without you needing to do anything. Please refer to the table below for more details about the different components in the study.

5.1. Return consent form

If you are interested in the RADAR Sleep Hygiene Study, we will ask you to complete and return the consent form.

5.2. A handbook to setup the smartphone apps

If you decide to participate, we will send you a handbook providing you with details on how to install and setup both of the apps. This will take approximately **20 minutes**.

5.3. The Participation Period

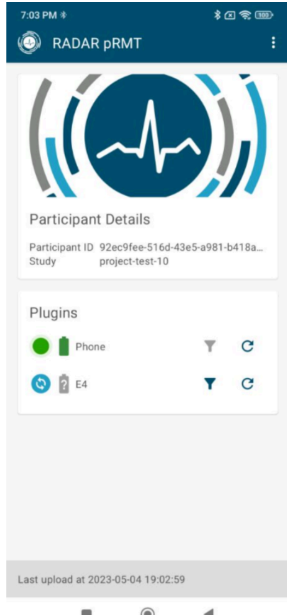
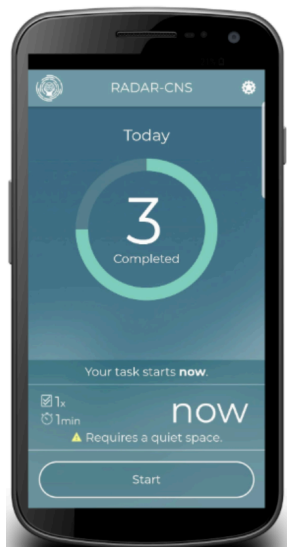
Once you are set up, we will invite you to wear your wearable device for as much of the time as possible over the participation period, including at night. The smartphone apps will guide you on what else to do. This will involve receiving notifications or reminders on your smartphone to answer some short questions. These will usually take about 3 minutes, but on some days might take up to 10 minutes.

During the study, we ask you to go about your daily life as usual, and to keep your smartphone charged (we recommend you charge it each night or whenever necessary, to keep it on). You will be given the contact information for the research team in case you have any problems. If we stop receiving information from the apps or the wearable device, we will contact you to check that they are still working and to troubleshoot any problems, but we will not be monitoring the data in real time and will not be reacting to content submitted.

At the end of three months, you'll have the option to continue sending us data for as long as you are happy to, up to a year. This is to look at sleep patterns over time. This continuation is optional, and you can choose to stop at any time.

6. Description of Data Collection

	The Wearable Devices
	These devices are commercially available fitness trackers, which passively collect information about your heart rate and activity levels . They send this information to the smartphone

	using wireless technology. From this information, we will be able to calculate levels of physical activity and sleep patterns. The devices that are currently incorporated with RADAR Base platform are Fitbits, Oura Rings, Garmin device, E4 Empatica, Pebble 2 and Polar device.
	The Passive App The RADAR passive app collects information about smartphone usage (apps, call and SMS logs and number of contacts); environment (light levels and Bluetooth devices); activity and travel (relative location (see below) and activity sensors) and other smartphone data (device information, battery and power station and record counts and server status). All the information is de-identified. This means, for example, we won't record your real-world location, but instead, a relative location provided by offsetting coordinates by a random amount. We also don't have access to the content, name or contact details of the people you call or message. You can choose what information the passive app has access to using permission settings on your smartphone.
	The RADAR Active App The RADAR app will send multiple notifications for you to complete sleep quality, mood questions every morning, daytime sleepiness question every afternoon, concentration questions twice a week, social communication and depression questions each fortnight, and a monthly sleep routines and phone usage questionnaire. For a separate set of questionnaires, you will have different deadlines within which you may complete the questions.
	Cognitron Cognitron is a flexible platform for assessing cognition and mental health remotely via web browsers and mobile applications. More details about Cognitron can be found in the link. https://www.cognitron.co.uk/

The apps will continuously send smartphone information to us, using a wi-fi connection. We will **not** be able to access any other information, such as the content of calls or messages or what websites are visited.

7. Are there any risks or disadvantages of taking part?

The RADAR Sleep Hygiene study will take up a short amount of time each day in order to respond to notifications on your smartphone. You are welcome to skip any questions that you'd prefer not to answer. We also ask that you do not complete the questions on the smartphone apps when in any situation in which smartphone use is not allowed or inadvisable, e.g., driving, crossing the road, or at certain places of work.

There is a small risk that the technologies used in this study could be hacked, however this risk is the same as with any smartphone or commercial fitness tracker use. Encryption and data de-identification processes have been put in place to minimise any risk to you in the unlikely event of hacking.

You may find answering questions on the smartphone apps confusing or awkward. We hope this study will help us learn how to make them easier to use and will ask you about any difficulties at the end of the study. Some people may feel uncomfortable having their sleep or location information collected. We will make sure to explain exactly what data we are collecting, and how the data will be protected. You can also contact us at any time if you feel unhappy about any part of the study.

8. What are the possible benefits of taking part?

Taking part in this study will not benefit you directly. We know that in recent times, smartphones have become an inseparable part of people's lives and the potential of pathological or addictive use may exist in the population and across different demographic groups. This study will help us investigate the impact of smartphone use habits on sleep hygiene, that is, the impact of different types of phone interactions before bedtime on the quality, length, and effectiveness of sleep.

9. Data handling and confidentiality

Kings College London is the Sponsor of this research and is responsible for looking after your information.

We will use information from you for this research project. The research team and authorised administrative staff will use this information to do the research or to check your records to make sure that the research is being done properly.

Your data will be processed in compliance with UK data protection laws, including the UK GDPR and the Data Protection Act 2018.

9.1. Storage and access to identifiable information

Any identifiable data such as consent forms and contact details will be stored on a secure server within the university. The only people who will have access to information that identifies you will be the research team (who will use the information to contact you to check if you are getting on OK during your participation), and auditors (who may use it to check the research is being done properly). Study data will not be linked back to your name or contact details when it is analysed. No names will be used when the research is written up for publication; your contribution will be completely de-identified. All identifying information will be deleted upon study completion on [Date].

9.2. Pseudonymization

To maintain your privacy, your study data will be pseudonymised using a random unique ID. This means your name will be replaced with a code, and the information collected from the questionnaires, apps and Fitbit will only be associated with this code. The only link to your identity will be kept separately and used by the study team only to contact you during the study, provide you with your own data or to request its deletion.

9.3. Storage and usage of Pseudonymized data

King's College London has a responsibility to keep information collected about you safe and secure, and to ensure the integrity of research data. Specialist teams within King's College London continually assess and ensure that data is held in the most appropriate and secure way. This may include storage of anonymised or pseudonymised data with a contracted GDPR compliant third-party storage provider within the UK, where they are assessed as the best data storage option. In such cases the third-party storage provider will not have access to any data that could directly identify you.

10. Will my data be shared?

Sharing of scientific data among researchers helps to speed the rate of scientific progress enormously. We will therefore ask you whether you are willing for your de-identified data to be shared with other research groups or databases. Your data will be shared with other researchers around the world and used in future research projects after ethics approval. These projects can take place in universities, hospitals, non-profit groups, companies, and/or government laboratories. All researchers must respect the laws and ethical guidelines for biomedical research.

You can tell us in your consent form whether you do or do not wish to share your data beyond this project. This decision will not affect your participation in any way.

11. Data protection statement

The data controller for this project will be King's College London (KCL). The University will process your personal data for the purpose of the research outlined above. The legal basis for processing your personal data for research purposes under GDPR is a 'task in the public interest'. You can provide your consent for the use of your personal data in this study by completing the consent form that has been provided to you.

If you would like more information about how your data will be processed under the terms of UK data protection laws, please visit the link below:

<https://www.kcl.ac.uk/research/support/research-ethics/kings-college-london-statement-on-use-of-personal-data-in-research>

12. Will I get any feedback on my data?

Upon its conclusion, the findings of this study will be published in suitable journals and be listed in RADAR-Base website. We will also provide you with statistics regarding your own data summary.

13. Will I get paid for study participation?

There are no monetary incentives to participate in this study.

14. Can I change my mind after I have agreed to take part?

You are free to withdraw at any point of the study, without having to give a reason. Withdrawing from the study will not affect you in any way. You are able to withdraw your data from the study within the first 30 days of your participation, after which withdrawal of your data will no longer be possible because all identifying information will be destroyed. However, you will no longer be identifiable from the remaining data.

15. What if I have further questions, or if something goes wrong?

If you have a query or concern about any aspect of this study, please speak to the researchers who will do their best to answer your question. If they are unable to answer your questions or you have a complaint, please contact the principal investigator: [insert local investigator and email address]. You are also welcome to seek further advice from [insert local PALS name and phone number].

The following link is provided for more information on the RADAR-Base platform.

RADAR-Base: <https://radar-base.org>

Contact Information:

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